Real-Time UPI Fraud Detection Using Machine Learning

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Abstract— The rapid expansion of digital banking and mobile payment technologies has significantly transformed the financial ecosystem, particularly in India where the Unified Payments Interface (UPI) has emerged as one of the most widely used payment platforms. UPI allows users to perform instant fund transfers between bank accounts using mobile devices without the need to share sensitive banking information. The convenience, speed, and accessibility offered by UPI have encouraged millions of users to adopt cashless transactions for everyday financial activities such as bill payments, retail purchases, subscriptions, and peer- to-peer transfers. However, the increasing reliance on digital payments has simultaneously created new opportunities for cybercriminals to exploit users. Fraudsters increasingly employ social engineering methods such as phishing links, fake collect requests, fraudulent QR codes, impersonation phone calls, and malicious remote access applications to deceive users into authorizing unauthorized transactions. Traditional banking fraud detection systems are primarily rule-based and depend on predefined conditions such as transaction limits or suspicious locations. These approaches are ineffective in identifying modern behavioral fraud because attackers often imitate normal user activity patterns to bypass detection mechanisms. This research proposes a real-time UPI fraud detection system based on machine learning techniques that analyze transaction behavior dynamically rather than relying on static rules. The system monitors transaction parameters including transaction amount, frequency of transactions, time pattern, device identification, geographical location, and user behavioral history. Supervised learning models such as Logistic Regression, Random Forest, and Gradient Boosting are trained on historical transaction datasets to classify transactions as genuine or fraudulent. A risk score is generated for each transaction, and if suspicious activity is detected, the system immediately sends alerts to both the user and the banking authority to prevent financial loss.

Index Terms—UPI security, fraud detection, machine learning, digital payments, cybersecurity, behavioral analysis.

I. Introduction

Digital payment systems are integral to today's banking system with the introduction of Unified Payments Interface (UPI) that facilitates instant money transfer through mobile phone. While providing the ease of conducting transactions with the portal available anytime and from anywhere, it has made online activities like bill, pay, shopping, money transfer, etc, ubiquitous. This increased usage has also increased the number of cyber frauds wherein the attacker can deceive the user by sending phishing messages, false payment requests, social engineering techniques, etc. Using rule-based security practices like transaction limits, location limits, etc, cannot identify new and evolving fraud.

Machine learning is a better approach as it does transaction behaviour analysis and detects abnormal behaviour of the transaction. Every user has a normal pattern of transaction based on the amount spent, time and device usage and any inconsistency can be considered as a fraud transaction. This project proposes a real-time UPI fraud detection platform which does continuous monitoring of the transaction and classifies as real or fraud. By bringing together behavioural monitoring with real-time alert notifications, fraud transaction can be prevented.

II. Related Work

A large portion of the earlier research on financial fraud detection was on credit card transactions with focus on statistical analysis and simpler classification algorithms. Prior to deployment, the earlier algorithms used simple rule, checks and anomaly detection to check for irregularities. These methods were capable of only prediction of basic anomalies with very high false alarm rates. Later on, research was focused on various machine learning models such as Decision Trees, SVMs, and ensemble models to provide greater prediction accuracy by understanding patterns in transactions and user behaviour.

Most of these solutions were designed for processing credit card transactions offline and not in real-time as had become necessary with UPI based transactions. Unlike credit card transactions, UPI frauds usually involved social engineering and required user authorisation, hence behaviour analysis was more necessary than fixed limits on a transaction threshold. A real time system capable of monitoring user behaviour based on machine learning and instantly notifying users on their mobile devices would be an ideal solution. The proposed research addresses the same.

III. Methodology

3.1 Data Collection Module

The data collection module collects the data of transactions which is needed for fraud detection. It collects the history records of UPI transaction such as transaction amount, time, date of the payment, location, device id, number of transactions in a time period etc. It also takes data about the previous user transaction behavior into account. These records are then used as training data for machine learning algorithms to detect real and fake transactions.

3.2 Data Preprocessing Module

The raw transactional data is not ready for machine learning analysis, as it has missing values and duplicate data, irregular formats cannot be used directly. The preprocessing components eliminates duplicated data and organizes data set into structured format; process categorical parameters (e. g. Device type, location), encoded them to numerical value for machine learning algorithms; normalization of features and attributes are used in the process to make sure all the features maintain similar range and scales. This stage guarantees a robust input data to prediction engine.

3.3 Feature Extraction Module

The feature extraction module takes the cleaned transaction data and turns it into indicators of user behavior. It looks for patterns in the way users operate, for example, in their typical amount of money, time of day, location, or device used. Each anomaly in these measures, such as a large payment, a transaction at a strange hour, or a payment from a new device, is considered a red flag indicator. These behavioral features allow the system to detect abnormal activity more accurately than just using raw transaction details.

3.4 Machine Learning Model Training Module

The dataset prepared in this module is split into training data and testing data in order to test the efficacy of the models. The supervised algorithms Logistic Regression, Random Forest, and Gradient Boosting are trained based on the past transaction data. They are trained to learn the patterns of original transactions as well as the fraudulent transactions, with the purpose of developing a model to determine frauds. Performance measures including accuracy and recall are utilized to determine the performance of each of the models and the most accurate one is used to detect in real time.

3.5 Fraud Detection Module

The fraud detection module runs in real-time on the admission of live transactions. On every attempt of payment made by a user, system extracts transaction information and feeds the trained model with these features. The model predicts a score which is a 'Likelihood of Fraud' and compares it against a threshold value. It then signs the transaction as valid or suspect. This module facilitates real, time financial fraud monitoring.

3.6 Alert and Notification Module

Once the malicious transaction is identified, the alert & notification module immediately notify the user and notifies the banking authority for taking commercial action. The notification at the user end helps to cross verify the transaction and at bank end can either block the payment or even give additional customer authentication process. This type of system of quick response is very helpful to avoid financial loss & increase user awareness on malicious act.

3.7 Reporting and Monitoring Module:

Report and monitoring module saves all kind of information about transactions and prediction results in a remote database. The saved database is used to generate fraud and trend reports on transactions in order to let the bank and administrator control security measures and trends over frauds. The stored data will be used periodically to retrain the machine learning model so the system could evolve with the new fraud techniques.

IV. Experimental Results

The experimental results proved that potential fraudulent transactions can be detected by the proposed system by exploring the various aspects of fraudulent transactions such as time difference between the current and last transaction, behaviour pattern such as no. of transactions on a certain day/time, use of device or location. Many frauds were discovered even when the amount involved in it was less and transaction limits are not being breached. The proposal was able to detect various types of frauds such as multiple fast transactions, use of new device and customer payments at odd hours

The performance evaluation of machine learning models also showed that the ensemble methods can deliver a performance higher than that of simple statistical classification. Among all machine learning models, Random Forest provided the most stable performance, for it used and combined multiple behavioural features, and Logistic Regression was best suited for basic statistical classification, while Gradient Boosting enabled detection for more delicate fraudulent signals. Alert notification method allowed users and banking institution to react in a timely manner, greatly helped in information security and risk management, and optimistic users.

V. Fraud Detection Algorithm

1. The process starts when the user initiates a transaction through the UPI mobile application.
2. The system logs transaction data as transaction amount, timestamp, device data and customer/merchant location.
3. The transactions data is sent to the preprocessing module where cleaning, encoding and normalization are done.
4. The feature extraction section. It comprises of analyzing user most common way of behavior such as day to day spend pattern, frequency of transactions and abnormal behavior.
5. The processed features are provided to the machine learning prediction engine.
6. The trained model evaluates the transaction and generates a fraud risk score.
7. The decision module compares the risk score with a predefined threshold value.
8. If the risk score is low, the transaction is approved and forwarded for bank processing.
9. If the risk score is high, it sounds an alarm to the user and the bank, and holds or authenticates the transaction.
10. The transaction result (genuine or fraudulent) is stored in the database and the process ends.

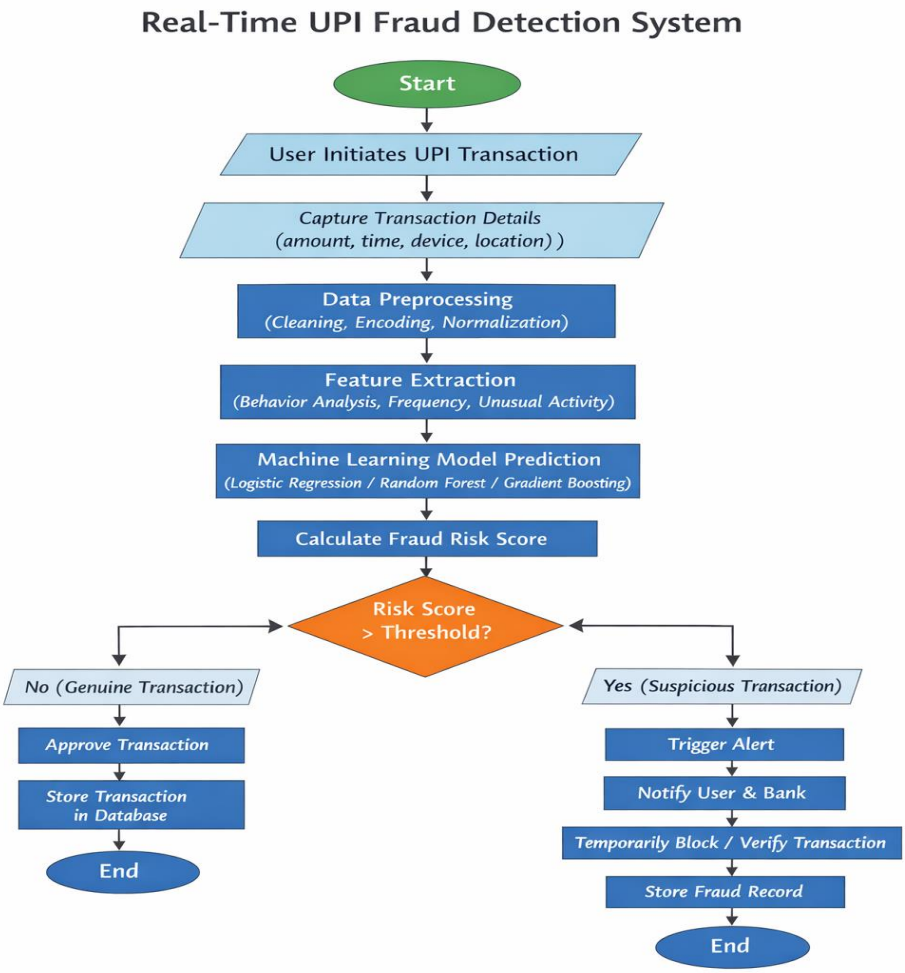

Fig. 1. Workflow of real-time UPI fraud detection using machine learning.

VI. Application of the System

The real-time UPI fraud detection system proposed here can be integrated with the banking application, digital wallets and other financial technology (FinTech) platforms to improve security of transactions and increase the confidence of all users. This real-time monitoring system can detect any suspicious activity and alert the concerned authorities instantaneously resulting in preventing any fraudulent transfer of funds from customer's account. The real-time alert service can also trigger the user to validate their transactions immediately and avoid any undesired transaction order. It also minimizes the manual validation process by bank officials and adds to the business efficiency. Further, this detection framework can be implemented for other forms of digital payments too like credit cards, internet banking etc. Making it a useful integrated solution.

VII. System Architecture

- User Mobile Application
- UPI Transaction Server
- Data Preprocessing Module
- Feature Extraction Module
- Machine Learning Prediction Engine

- Fraud Decision Module
- Bank Processing Unit
- Alert and Notification System
- Database Storage

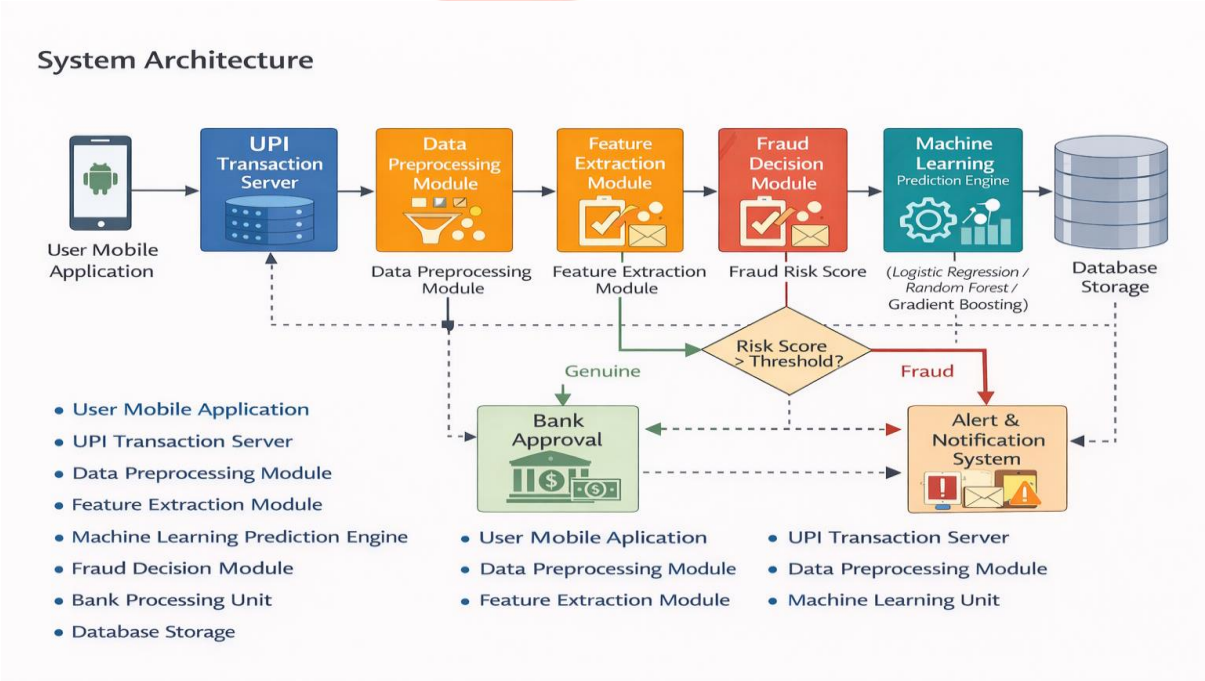

Fig. 2. System architecture of the real-time UPI fraud detection system.

VIII. Results and Discussion

The real-time UPI fraud detection system suggested in here can be used to provide more secure transactions for the banking application, digital wallets and other FinTech platforms thus providing more assurance to the end users. With the help of this real-time monitoring system, any malicious activity can be detected and notified to the corresponding authorities immediately thus helping in preventing any malicious transfer of money from the customer account. This as well will immediately generate an alert to validate the transactions with the users order which can be redrafted avoiding any unwanted transfer of permissions to transfer. This will also reduce the cross-validation task for the bank officials helping in improving business efficiency. This concept can be extended for any other type of digital transactions like credit card payments, internet banking etc.

IX. Research Limitations

The system is trained on existing transaction data available; performance of system is a function of dataset quality and diversity (the inclusion of various types of offenders). The system is restricted to behavioural features only and does not model all the possible frauds present in real world. Prediction accuracy and adaptability to new types of frauds will increase by the use of larger real time datasets and regular update of models.

X. Conclusion

This study proposes a real-time UPI fraud detection system using machine learning algorithms which addresses a significant improvement of digital payment transactions security. The payment transaction behaviour, like transaction time, location attributes, device information, transaction amount, spending behaviour etc. was also examined to recognize the suspicious behaviours before the execution of the transaction. Application of supervised learning methods helps to accurately classify either of the transactions as ‘fraudulent or genuine’ and at the same time raises fewer false alarms. At the same time;

the alert mechanism was based on real-time notification to both the customer and bank for further measures to be taken.

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XII. References

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